

Prototype Validation of a Large-Scale CO₂-to-Formate Zero-Gap Electrolyzer

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The scale-up of gas-phase CO₂ electroreduction to formate is crucial for its industrial application but remains largely unexplored. This work presents the design and validation of a 100 cm² electrolyzer prototype featuring a zero-gap configuration and a serpentine flow field to ensure uniform CO₂ distribution. Scaling up a CO₂ electrolyzer requires optimized flow field design, in this case, a serpentine geometry enhances CO₂ transport and minimizes mass transfer limitations, thereby improving overall performance. Experimental prototype testing is conducted to evaluate the effects of current density and water content in the CO₂ feed. Optimal performance is achieved at 200 mA cm⁻² and

a water content of 15 g h⁻¹, yielding a formate concentration of 760 g L⁻¹, a Faradaic efficiency of 67%, a production rate of 7 mmol m⁻² s⁻¹, and an energy consumption of 507 kWh kmol⁻¹. Comparisons with a 10 cm² lab-scale reactor reveal improved CO₂ conversion and production rate, validating the benefits of optimized flow field design and scale-up approach. While energy efficiency is somewhat reduced to increased Ohmic losses, the overall results support the technical feasibility of scaling gas-phase CO₂-to-formate electrolysis. Further improvements in design and energy management are still needed to advance toward industrial implementation.

1. Introduction

The increasing concentration of atmospheric CO₂ is a major contributor to climate change. While efforts such as renewable energy deployment and improved energy efficiency (EE) have advanced, certain industrial processes, particularly in the cement, steel, and chemical sectors, remain difficult to decarbonize due to the inherent nature of their CO₂ emissions.^[1–3] In this context, Carbon Capture, Utilization, and Storage (CCUS) technologies have gained attention as a viable strategy to reduce emissions by capturing CO₂ and converting it into valuable carbon-based products within a circular economy framework.^[4,5]

Among CCUS approaches, the electrochemical reduction of CO₂ stands out for its flexibility and operation under mild conditions using electricity as the energy input.^[6] This process can selectively produce compounds such as CO, CH₄, C₂H₄, CH₃OH,

and formate (HCOO⁻), with the latter being particularly attractive due to its potential as a hydrogen carrier and industrial feedstock.^[7–9] Catalysts based on metals like Sn, Cu, Pd, and Bi have been developed for this purpose, with bismuth-based materials showing excellent selectivity, efficiency, and long-term stability in the electroreduction of CO₂ to formate.^[10,11]

A crucial factor in the success of CO₂ electroreduction to formate is the design and configuration of the electrochemical reactor.^[12] Initial investigations commonly employ H-type cells, which are simple to assemble and useful for evaluating catalyst performance. These systems consist of separate cathode and anode compartments, divided by an ion-exchange membrane, with electrodes immersed in a liquid electrolyte saturated with CO₂ gas.^[13] However, such setups suffer from low reaction rates and dilute product concentrations, limiting their potential for scale-up.

To overcome these limitations, flow cells have been developed as a more advanced and scalable reactor design.^[14] These cells feature separate anodic and cathodic compartments, divided by an ion-exchange membrane, which ensures ionic conductivity while preventing the reoxidation of formate at the anode. Unlike H-type cells, flow cells operate with a continuous electrolyte feed, and CO₂ is supplied in the gas phase independently of the electrolyte.^[15] This configuration significantly increases the availability of CO₂ at the catalyst interface, thereby enhancing reaction performance and making flow cells a more viable option for industrial-scale implementation.

A critical component of these systems is the gas diffusion electrode (GDE), which facilitates CO₂ transport through its porous structure to the catalyst surface.^[16] The reaction occurs at the triple-phase boundary (gas–liquid–solid), where CO₂, electrolyte, and catalyst meet. While some designs use a liquid

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electrolyte between the membrane and GDE, this often results in a product dilution.^[17] To address this challenge, an effective strategy is to place the GDE in direct contact with the ion-exchange membrane, thereby eliminating the need for a liquid catholyte. In this configuration, a humidified CO₂ stream is supplied to sustain the reaction, enabling the production of formate at concentrations exceeding 300 g L⁻¹.^[18] **Figure 1** shows a comparison of the operation of these two configurations. In the gas-phase configuration (Figure 1a), the GDE is placed in direct contact with the ion-exchange membrane, forming the zero-gap cathode, and both the unreacted CO₂ and reduction products are extracted through the gas flow channel. In contrast, in the liquid-phase configuration (Figure 1b), a liquid catholyte compartment is introduced between the GDE and the membrane, where liquid products are collected. In a flow-through system, the unreacted CO₂ exits through the catholyte compartment along with the liquid solution.

At the laboratory scale, flow cells have demonstrated high activity in the electroreduction of CO₂ to formate, both in systems using liquid electrolytes and in zero-gap configurations without them.^[10] Several studies reported in recent years at laboratory scale have demonstrated high Faradaic efficiencies (FEs) and significant formate production rates, supporting the feasibility of scaling up the CO₂ electroreduction to formate in a gas-phase configuration. In this context, the most relevant works from the literature have been compiled in **Table 1**, which presents the types of cathode, anode, and anolyte used, highlighting the FEs achieved.

Moreover, small-scale studies have facilitated the identification and analysis of the primary deactivation mechanisms that limit long-term performance.^[19] One of the most critical issues is salt precipitation within the porous structure of the GDE,

which can obstruct CO₂ transport to the catalyst surface. Another common degradation pathway is GDE flooding, which typically arises over extended operation times when the electrode's hydrophobicity is compromised. This allows electrolyte intrusion into the gas channel, impeding CO₂ transport and favoring the competing hydrogen evolution reaction (HER).

Recent research has explored several strategies to mitigate these deactivation phenomena. For salt precipitation, the substitution of conventional alkali cations with more soluble species, such as Cs⁺^[20] or the use of acidic electrolytes,^[21] has shown promise in reducing salt accumulation within the GDE. Regarding the flooding, studies emphasize the importance of optimizing the electrode composition to preserve hydrophobicity and maintain stable performance over time.^[22] Together, these insights provide a foundation for extending the operational lifetime of CO₂ to formate electrolyzers and enhancing their scalability potential.

Building on this foundation, the development of larger-scale flow cells is a crucial step forward, translating laboratory findings into practical applications.^[23] While typical lab-scale devices operate with electrode areas on the order of a few cm², industrially relevant prototypes must scale up to at least 100 cm² to enable meaningful performance validation under more realistic conditions. However, this upscaling introduces engineering challenges that are often negligible at the laboratory scale.^[24] The first of these is the design and fabrication of the electrode with a larger geometric area. For this purpose, the use of automated fabrication through the spray pyrolysis technique facilitates the automation of its production, ensuring reproducibility in the manufacturing of large-scale electrodes.^[25] In this regard, the use of a GDE in a zero-gap configuration within the system will also present additional challenges related to the distribution of CO₂ over its surface and the pressure drops that may occur within the flow field. Achieving uniform CO₂ distribution across the GDE surface is essential to avoid performance-limiting effects such as concentration gradients, dead zones, or localized velocity peaks.^[26,27]

In addition to flow management, pressure balancing across the GDE becomes increasingly important as systems are scaled up. Although often overlooked in small-scale configurations, pressure gradients can have a pronounced impact on both stability and performance. Fink et al.^[28] illustrated several failure scenarios resulting from pressure imbalance: insufficient CO₂ pressure relative to the liquid side can lead to electrolyte flooding of the GDE, obstructing gas transport; conversely, excessive CO₂ pressure can cause oversaturation, bubble formation, or CO₂ losses via carbonate formation.^[29] Thus, maintaining a well-controlled pressure differential across the GDE is essential to prevent performance degradation and ensure operational stability, with these effects being more pronounced in systems that use a liquid catholyte.

Despite the progress made, several challenges remain in scaling up CO₂ electroreduction systems. These include low carbon conversion efficiencies, HER competition, high energy demands, mass transfer limitations, and the absence of efficient, scalable components, and reactor architectures.^[30] Nevertheless, recent developments have demonstrated significant progress toward overcoming these obstacles.^[31] A notable example is the system developed by Fink et al.,^[28] featuring a stacked electrolyzer with

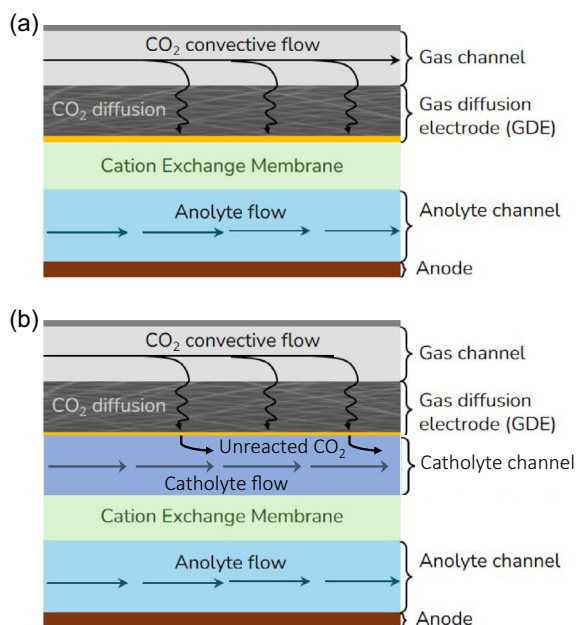


Figure 1. Schematic representation of the electrolyte operation for a) gas-phase configuration and b) liquid-phase configuration.

Table 1. Main figures of merit for the CO₂ electroreduction to formate in gas phase reported in the literature.

Cathode	Anode	Anolyte	[HCOO ⁻] [g L ⁻¹]	j [mA cm ⁻²]	FE [%]	EC [kWh kmol ⁻¹]	[Ref.]
Sn	Pt/C	1 M KOH	116.0	40	78	152	[49]
Sn	DSA	1M KOH	19.2	45	49	244	[50]
Sn	Ni Foam	2 M KOH	65.4	100	80	181	[51]
Bi	DSA	1 M KOH	43.2	200	43	434	[52]
Bi	DSA	1 M KOH	337.0	45	89	180	[18]
Bi	DSA	1M KOH	312.0	200	25	547	[18]
Bi	DSA	1 M KOH	10.8	600	74	342	[40]
Bi-Sn	IrO _x on Ti Foam	0.1 M KHCO ₃	153.0	60	82	267	[53]
Bi-Sn	IrO ₂ on Ti Foam	0.1 M KHCO ₃	0.36	120	80	268	[54]
Bi	DSA	0.3 M K ₂ SO ₄ (pH 1)	88.0	200	42	358	[21]
Bi	CaTiO ₃ -BiVO ₄	0.5 M KHCO ₃	63.8	32	79	318	[55]
Bi	Ni-Co Foam	1 M KOH + 1 M glycerol	359.0	45	95	192	[56]
Bi	NiFe Foam	Textile industry stream (NaOH)	260.0	200	62	380	[57]

an active area of 3052 cm², which achieved FE of 61% for potassium formate production over 2 h using a, Sn-based catalyst. Similarly, OCOChem's commercial Carbon Flux Electrolyzer has demonstrated the feasibility of large-scale CO₂-to-formate conversion under ambient conditions, yielding a high-purity product.^[32]

However, all existing scaled systems reported in the literature have employed liquid electrolyte configurations, either in single cells or in stack arrangements, where scaling is typically achieved by expanding the active electrode area. To date, no prototype operating at scale has incorporated a zero-gap cathode configuration, an architecture that offers potential advantages in terms of compactness, ionic conductivity, and EE.

To address this critical gap, the present work proposes a novel approach: the development of a 100 cm² zero-gap cathode prototype for CO₂ electroreduction to formate. This effort covers the full development chain, starting with the design of the electrolyzer, electrode scale-up and fabrication, and experimental validation. The GDE, based on a bismuth catalyst, is fabricated via a previously optimized automated spray pyrolysis technique,^[25] enabling reproducible and scalable electrode production. The complete prototype is tested under continuous operation, and performance will be evaluated through key metrics such as FE, energy consumption (EC), EE, and single-pass conversion efficiency (SPCE), under a range of operating conditions. This integrated approach aims to demonstrate the feasibility and advantages of zero-gap configurations at relevant scales and to contribute to the advancement of scalable and efficient CO₂ electroreduction technologies.

2. Experimental Section

2.1. GDE Fabrication

The GDE, serving as the cathode in the electrolyzer, features an active geometric area of 100 cm². It is composed of a carbon

paper substrate, specifically AvCarb MGL 190 (50% wt PTFE-treated, FuelCell), treated with 50% PTFE to enhance hydrophobicity. The catalytic layer was deposited onto this substrate via an automated spray pyrolysis technique.

The catalytic ink was formulated by dispersing 150 mg of commercial Bi₂O₃ (particle size 90–210 nm, Sigma-Aldrich), 150 mg of Vulcan XC-72 R (Cabot), 1280 mg of Nafion D-521 solution (5 wt% in water and 1-propanol, with an exchange capacity of ≥0.92 meq g⁻¹, Ion Power), and 325 mg of PTFE (polytetrafluoroethylene preparation, 60 wt% dispersion in H₂O, Sigma-Aldrich) in 25 mL of ethanol (Ethanol, 96% v/v pure, pharmaceutical grade, PanReac AppliChem). The mixture was homogenized via ultrasonication.

The GDE was fabricated using a previously optimized automated spray pyrolysis technique,^[25] ND-SP Mini Ultrasonic Spray Coater (Nadetech Innovations), ensuring uniform deposition over the 100 cm² area and reproducible electrode production. These GDEs, prepared with the same composition and fabrication technique, were characterized in previous studies.^[21]

2.2. Electrolyzer Design and Construction

A custom-built, two-compartment zero-gap electrolyzer was designed and manufactured by Apria Systems S.L. The reactor featured a gas-fed cathode configuration in which the Bi-based GDE was placed directly against a cation-exchange membrane (CEM), specifically Nafion 117 (Ion Power).

The system was secured between two AISI 316 stainless steel tightening plates, each with 12 bolts to ensure uniform compression and gas-tight sealing. Internally, the cathode and anode compartments were machined from polypropylene (PP), each with a dedicated inlet and outlet. A serpentine flow field was integrated into the cathodic PP plate to ensure homogeneous CO₂ distribution.

Viton gaskets were used throughout the assembly to prevent leakage. A separate chamber facilitates anolyte circulation between the anode and membrane. The anode was a custom-fabricated titanium plate coated with an Ir-MMO (iridium mixed metal oxide) via electrodeposition, enabling the OER. The cathode current collector consisted of a titanium plate with a 100 cm² central aperture, allowing direct exposure of the GDE to the CO₂ stream.

The electrolyzer had overall dimensions of 180 × 180 mm. Figure 2 presents a 3D rendering of the assembled system and its components.

2.3. Experimental Setup

The core of the experimental system was a 100 cm² electrolyzer. Humidified CO₂ was supplied to the cathode at 2 L min⁻¹ flow rate. Humidification was achieved using a vapor delivery module (Iberfluid), consisting of a CO₂ flow controller, a pressurized deionized water tank, and a control evaporator mixer (CEM Evaporator, Bronkhorst). The water content in the feed was adjusted to 7, 15, and 20 g L⁻¹ of water in the 2 L min⁻¹ CO₂ stream.

The anode was fed with a 1 M KOH solution (Potassium hydroxide, 85% purity, pharmaceutical grade, PanReac AppliChem), delivered at 0.57 mL min⁻¹ cm⁻², using a peristaltic pump (HF-ST, Hygiaplex). Electrochemical control was provided by a potentiostat-galvanostat (Arbin). The system operated at four current densities: 30, 100, 200, and 300 mA cm⁻². The operating conditions were set to reproduce reaction conditions similar to those used in previous studies carried out at the 10 cm² laboratory scale.^[18]

Each experiment was conducted in single-pass continuous flow mode over 2 h, with product collection and subsequent analysis. All conditions were tested in duplicate to ensure reproducibility and performance evaluation.

Formate production was quantified by ion chromatography (Dionex ICS 1100), employing an AS9-HC column and a 4.5 mM sodium carbonate eluent at a flow rate of 1 mL min⁻¹. Electrolyzer performance was evaluated based on key figures of merit, presented in the Supporting Information.

2.4. Figures of Merit

FE quantifies the proportion of the total current density supplied to the electrochemical cell that is effectively used for formate production.

$$FE(\%) = \frac{z \cdot M \cdot F}{j \cdot A \cdot t} \times 100 \quad (1)$$

where z is the number of electrons transferred in the reduction reaction (2 for formate), M is the number of moles of formate produced during the experiment, F is the Faraday constant (96 485 C mol⁻¹), j is the applied current density, A is the geometric active area of the electrode, and t is the experiment duration.

Formate production rate (r) evaluates the rate of formate generation per unit time and electrode area

$$r \left(\frac{\text{mmol}}{\text{m}^2 \text{s}} \right) = \frac{M}{t \cdot A} \quad (2)$$

where M , A , and t are as defined above.

EC represents the total electrical energy required to produce one kilomole of formate.

where V is the applied cell potential, and the other parameters are as previously defined.

$$EC \left(\frac{\text{kWh}}{\text{kmol}} \right) = \frac{j \cdot A \cdot V \cdot t}{M} \quad (3)$$

EE is portion of the total energy supplied that is harvested in the CO₂ reduction to formate.

$$EE(\%) = FE \cdot \frac{E_T}{E_{\text{Cell}}} \quad (4)$$

where FE is the FE to formate (%), E_T is the theoretical cell potential for the CO₂ reduction to formate in the cathode, coupled with the OER in the anode, and E_{Cell} is the overall voltage supplied to the cell.

SPCE assesses the fraction of CO₂ converted to formate in a single pass through the electrolyzer.

$$SPCE(\%) = \frac{j \cdot A \cdot t \cdot FE}{z \cdot F} \cdot \frac{V_{\text{mol}}}{V_{\text{IN}} \cdot t} \quad (5)$$

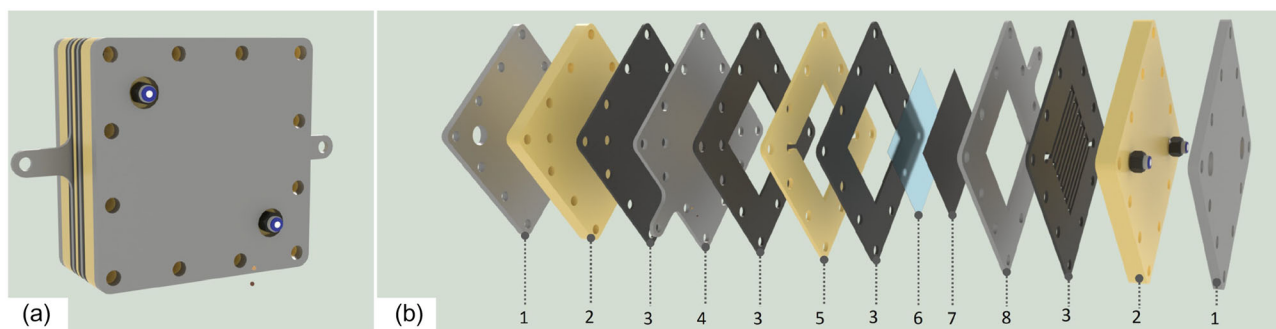


Figure 2. Rendered 3D image of a) the complete electrolyzer and b) internal components: 1) external tightening plates, 2) PP plates, 3) Viton gaskets, 4) Ir-MMO anode, 5) Anolyte distributor, 6) CEM, 7) GDE, and 8) Ti current collector.

Where V_{mol} is the molar volume of an ideal gas under standard conditions (22.4 L mol^{-1}), V_{IN} is the volumetric flow rate of CO_2 fed to the cell (L min^{-1}), and the remaining variables are as previously defined.

3. Results and Discussion

3.1. CO_2 Flow Field Geometry Design

The scale-up of the CO_2 electrolyzer from laboratory scale to a 100 cm^2 prototype presents several challenges, among which the management of CO_2 flow over the GDE surface is particularly critical. In this regard, a design that effectively addresses this aspect can significantly improve the overall performance of the electrolyzer.

To maximize the CO_2 access to the reaction zone, it is necessary to evaluate the relationship between the GDE surface area and the effective flow volume, defined as the portion of the flow field where the CO_2 stream circulates without dead zones or bypass areas. In such regions, the absence of convective CO_2 flow limits the potential for diffusive transport toward the active reaction site. In laboratory-scale filter-press reactors (10 cm^2), flow fields without any specific geometric design have been commonly used (Figure 3a). This can lead to the formation of dead zones, reducing the effective flow volume-to-GDE surface area ratio and hindering CO_2 mass transfer to the reaction zone.

To overcome these limitations, a serpentine flow field geometry is employed (Figure 3b). As shown in the CFD simulations in Figure S1, Supporting Information, utilizing a serpentine design instead of a nonstructured flow field helps prevent the formation of dead zones, thereby maximizing the effective volume of CO_2 in contact with the electrode surface. Based on the CFD simulation results, it is evident that in the nonconstructed flow field geometry, $\approx 75\%$ of the total volume exhibits near-zero flow velocity,

resulting in a significantly reduced effective volume. In contrast, the serpentine geometry shows low-velocity regions only in small areas adjacent to the channel walls, accounting for roughly 10% of the total volume as dead volume. These differences directly impact the ratio of effective volume to active area, which is 0.125 cm for the nonconstructed flow field, whereas for the serpentine geometry, this value doubles to 0.258 cm . Moreover, the serpentine configuration increases the fluid velocity, improving the Reynolds number from 310 to 780, which further enhances mass transfer conditions. This configuration promotes more effective CO_2 delivery, minimizing mass transfer resistance and improving reactant access to the catalytic site.^[33]

3.2. Prototype Experimental Validation

The 100 cm^2 CO_2 -to-formate electrolyzer prototype is experimentally evaluated to demonstrate and validate its operation, to prove the scalability of the gas-phase CO_2 electroreduction system. To this end, the electrolyzer is integrated into the experimental setup described above, where it is first commissioned to verify proper functionality. Subsequently, various operating parameters are investigated to better understand the system's behavior.

First, the influence of current density is assessed to determine the optimal operating range. Current densities of 30, 100, 200, and 300 mA cm^{-2} are applied while maintaining all other variables constant to ensure a consistent comparison. Specifically, the CO_2 flow rate is set at 2 L min^{-1} , with a water vapor feed rate of 15 g h^{-1} . The anode compartment is supplied with a 1 M KOH solution at a flow rate of $0.57 \text{ mL min}^{-1} \text{ cm}^{-2}$. The results regarding the effect of current density on formate production are presented in Figure 4.

As shown in Figure 4a, formate concentration increases linearly with current density between 30 and 200 mA cm^{-2} , reaching

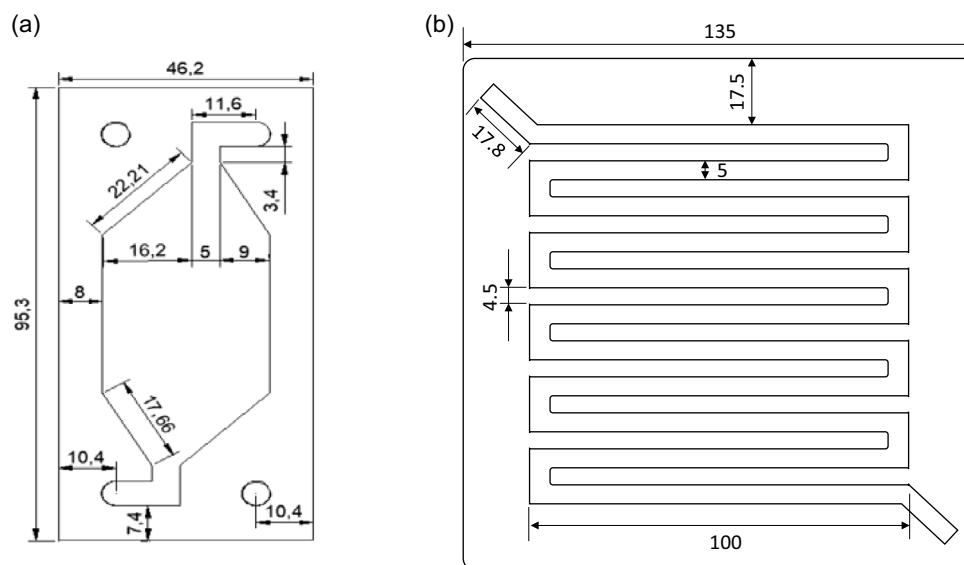


Figure 3. a) 10 cm^2 lab-scale flow field design, and b) 100 cm^2 proposed serpentine geometry for the flow field.

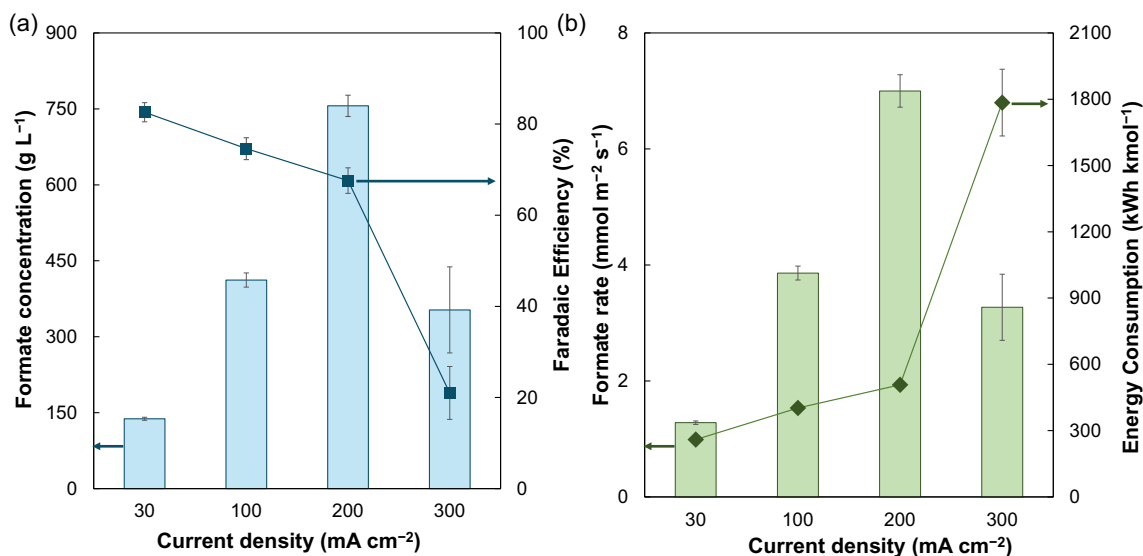


Figure 4. Results showing the influence of different current densities on a) formate concentration and FE and b) formate rate and EC.

a peak concentration of 760 g L⁻¹. However, at 300 mA cm⁻², formate concentration drops significantly to 353 g L⁻¹. This decline may be attributed to several factors. At higher current densities and large membrane areas, electro-osmotic water transport from the anode to the cathode can lead to GDE flooding.^[34] Flooding impedes CO₂ transport to the catalytic surface and promotes HER, a common deactivation mechanism for GDEs^[35–38]. Moreover, the crossover of alkaline electrolyte into the cathode can elevate the local pH, leading to CO₂ loss through non-Faradaic reactions such as carbonate and bicarbonate formation.^[39] These losses decrease the local CO₂ concentration near the catalyst, favoring HER over CO₂ reduction.

Regarding FE, a moderate decline from 82% to 67% is observed between 30 and 200 mA cm⁻², which is consistent with previous studies showing that higher current densities often lead to lower FE due to excess electron availability.^[40] This decrease can be attributed to mass transport limitations that lead to local CO₂ depletion,^[41] as well as possible catalyst transformations under more negative potentials that promote the HER.^[42] At 300 mA cm⁻², FE drops drastically to 21%, further confirming the dominance of parasitic reactions at higher current loads.

Figure 4.b shows the formate formation rate, which increases from 1.28 mmol m⁻² s⁻¹ at 30 mA cm⁻² to a maximum of 7 mmol m⁻² s⁻¹ at 200 mA cm⁻². This improvement results from the greater availability of electrons, favoring formate generation. However, at 300 mA cm⁻², the rate decreases to 3.27 mmol m⁻² s⁻¹ due to the competing reactions such as the HER. In terms of EC, operation at 30 mA cm⁻² results in a low cell voltage (≈4 V), yielding an energy demand of 260 kWh kmol⁻¹ of formate. Although 200 mA cm⁻² enhances formate production, the increase in cell potential to 6.4 V raises the energy requirement to 507 kWh kmol⁻¹, while at 300 mA cm⁻², the cell potential continues to increase up to 7 V. These increases in cell potential are related to the rise in Ohmic resistance within the cell, which may be caused by two effects in the anode compartment. First, the

higher current density leads to increased O₂ production, resulting in greater bubble formation on the anode surface. This bubble coverage reduces the electrolyte's contact with the active surface area of the anode, increasing the overpotential required to maintain the applied current density.^[43] Second, heating of the system was observed at high current densities, such as 300 mA cm⁻². This temperature rise likely enhances bubble formation on the IrO₂ anode, further decreasing the active area in contact with the KOH solution and thus increasing the Ohmic resistance.^[44]

Further insights into the electrolyzer performance are provided by key figures of merit shown in Figure 5, including EE, SPCE, and total formate production. These indicators are critical to assess the viability of scale-up for industrial applications.

As shown in Figure 5.a, the highest EE (34.1%) is achieved at lower current densities, primarily due to favorable cell voltages and higher FE values. In terms of SPCE, an increase from 0.9% to 4.7% is observed between 45 and 200 mA cm⁻², followed by a decline to 2.2% at 300 mA cm⁻². This trend reflects the balance between enhanced electron availability and CO₂ conversion, although the low SPCE values indicate that CO₂ is supplied in excess under the current operating conditions.

On the other hand, Figure 5b shows that the maximum daily formate production (0.275 kg per day) is achieved at 200 mA cm⁻², providing a reference point for estimating the number of units or stacks required for industrial implementation.

3.3. Influence of the Water Content in the CO₂ Feed Stream

One of the key variables in the gas-phase electroreduction of CO₂ to formate is the amount of water vapor cofed with the CO₂ stream. The effect of varying this parameter between 7 and 20 g h⁻¹ on different figures of merit is presented in Figure 6.

As shown, the optimal results across all figures of merit are obtained with a water feed rate of 15 g h⁻¹. In terms of formate concentration (Figure 6a), the lowest and intermediate water

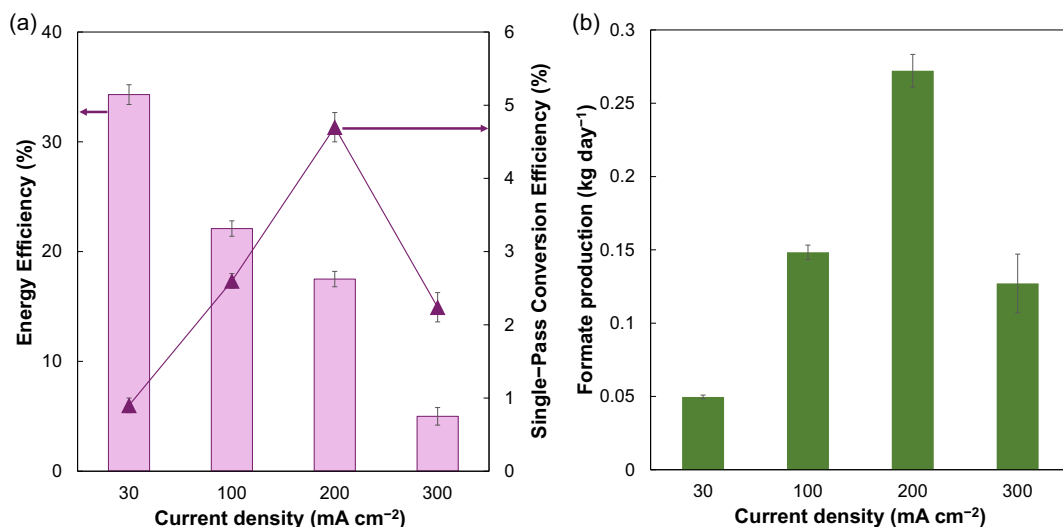


Figure 5. Results for the influence of different current densities on a) EE and SPCE and b) formate production.

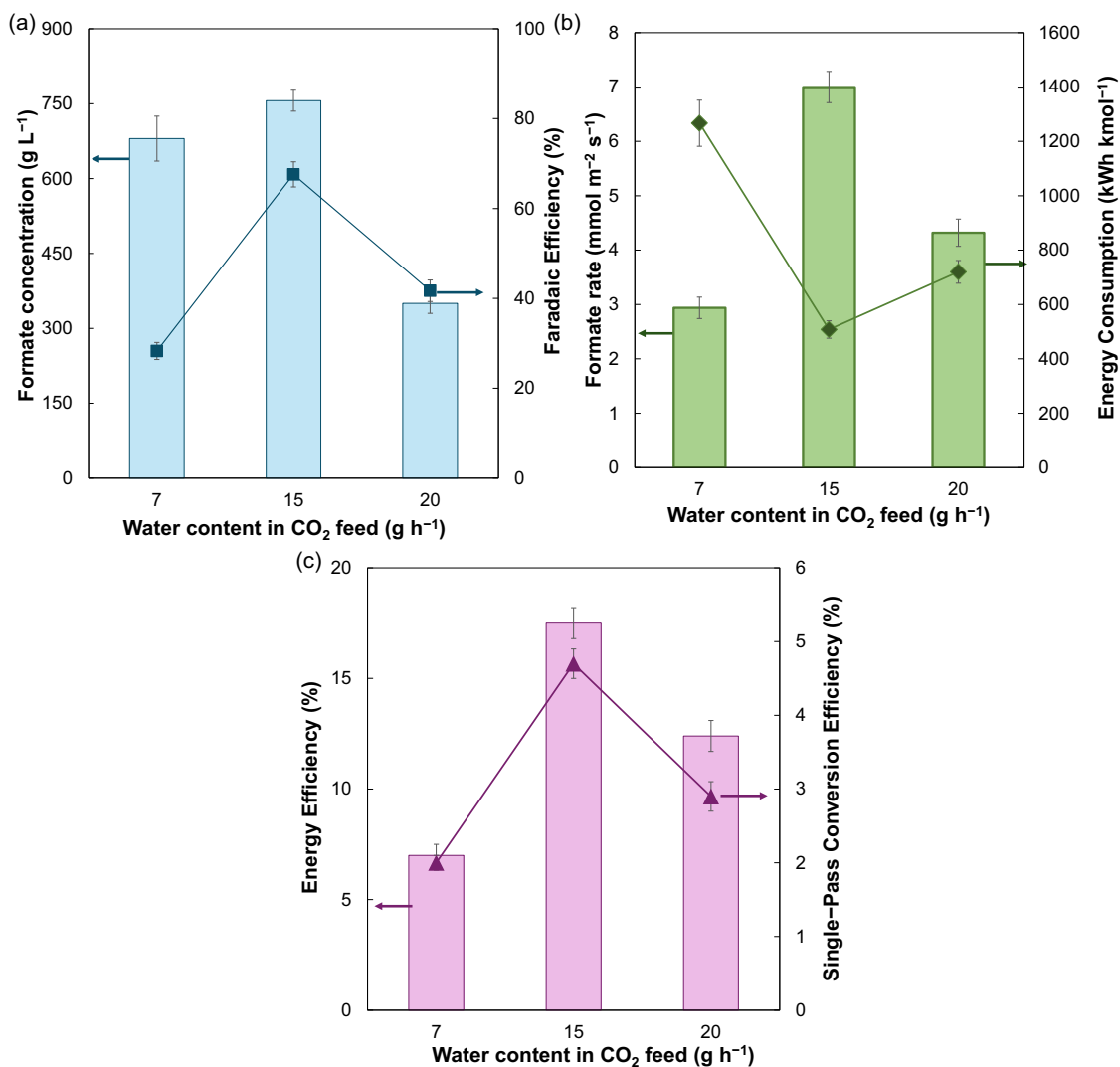


Figure 6. Experimental results showing the influence of different water contents in the CO₂ feed on a) formate concentration and FE, b) formate rate and EC, and c) EE and SPCE.

contents of 7 and 15 g h⁻¹ result in a more concentrated product, 680 and 756 g L⁻¹, respectively. However, at 7 g h⁻¹, this comes at the expense of a significantly lower FE (28.4%) compared to 67.5% and 41.6% at 15 and 20 g h⁻¹, respectively. A similar trend is observed for formate production rate and EC (Figure 6b), operation at 15 g h⁻¹ achieves both a high production rate of 7 mmol m⁻² s⁻¹ and the lowest EC of 507 kWh kmol⁻¹. In contrast, operating at 7 g h⁻¹ results in a markedly lower production rate of 2.94 mmol m⁻² s⁻¹ and significantly higher EC of 1266 kWh kmol⁻¹.

Insufficient water vapor in the feed stream can negatively affect CO₂ electroreduction performance. Primarily, a limited water supply may make water become the limiting reactant for the CO₂ reduction reaction, despite partial contribution from the anolyte.^[45] This leads to lower formate production rates and FE, as seen under 7 g h⁻¹ conditions. Additionally, low humidity can cause membrane drying, increasing the cell's internal resistance and elevating the required cell voltage,^[46] reaching 6.7 V in this case, thus demanding more external energy input.

On the other hand, excessive water vapor, specifically 20 g h⁻¹, also results in performance disadvantages. A higher water content dilutes the formate product and may lead to flooding of the GDE's porous structure, impeding CO₂ transport to the catalyst sites and promoting the HER.^[47] This is reflected in both the reduced formate production rate and FE. Supporting this, a drop in cell voltage from 6.4 to 5.6 V is observed, which may indicate partial flooding of the GDE. Therefore, 15 g h⁻¹ is identified as the optimal water feed rate, providing sufficient humidity without causing GDE flooding. This is corroborated by the highest values of EE and SPCE under these conditions: 17.5% and 4.7%, respectively (Figure 6c).

3.4. Lab Scale versus Prototype Performance

The transition from laboratory scale to a 100 cm² prototype development presents both challenges and opportunities to optimize

the electrolyzer configuration and improve the performance of CO₂ electroreduction to formate. In this context, the results obtained with the 100 cm² electrolyzer prototype developed in this work are compared with those previously reported for a 10 cm² lab-scale filter-press reactor.^[18] Both systems operate in a zero-gap configuration at the cathode, where the GDE is in direct contact with the ion-exchange membrane. The key performance metrics for both setups are summarized in Table 2.

The comparison of formate concentration highlights that the lab-scale reactor performs better at low current densities, achieving over 200 g L⁻¹ at 100 mA cm⁻², compared to 130 g L⁻¹ in the 100 cm² prototype. However, at 200 mA cm⁻², the prototype significantly outperforms the lab-scale reactor, reaching a formate concentration of 756 g L⁻¹ compared to 312 g L⁻¹. This improvement is attributed to the more effective distribution of CO₂ across the electrode surface, enabled by the serpentine flow field, which enhances mass transfer by increasing the effective CO₂ volume in contact with the electrode surface.

This prototype also exhibits a higher formate rate, with a peak of 7 mmol m⁻² s⁻¹ compared to 2.57 mmol m⁻² s⁻¹ for the lab-scale reactor. This is likely due to improved CO₂ accessibility to the catalyst surface. As a result, the SPCE improves significantly, from 1.7% to 4.7%, under optimal conditions at 200 mA cm⁻².

However, the increase in electrode area in the 100 cm² electrolyzer also leads to greater Ohmic resistance, reflected in a higher cell potential. While the lab-scale system maintains below 4 V, the prototype consistently exceeds this threshold, reaching 6.4 V at 200 mA cm⁻². Despite this, the EC remains comparable at 100 mA cm⁻² and is even lower in the prototype at 200 mA cm⁻² (507 kWh kmol⁻¹), primarily due to its superior formate productivity. Conversely, the EE is reduced in the prototype, likely due to energy losses as heat or increased anodic overpotential, which diverts energy from the CO₂-to-formate conversion.

In general, the new 100 cm² prototype demonstrates improved performance over the lab-scale reactor, primarily

Table 2. Comparative summary of the results obtained in the zero-gap configuration for the 10 cm² laboratory-scale electrochemical cell and the 100 cm² electrolyzer prototype.

Active Geometric Area [cm ²]	CO ₂ flowrate [L min ⁻¹]	Water inlet [g h ⁻¹]	Current density [mA cm ⁻²]	Cell Potential [V]	[HCOO ⁻] [g L ⁻¹]	FE [%]	Rate [mmol m ⁻² s ⁻¹]	EC [kWh kmol ⁻¹]	EE [%]	SPCE [%]
10	0.2	0.5	45	2.75	201	70.9	1.66	207.7	55.7	1.1
		1		2.9	158	61.3	1.43	253.4	42.8	1.0
		2		3	118	62.5	1.46	257.4	41.4	1.0
		0.5	100	3.1	256	40.6	2.1	409.1	23.2	1.4
		1		3.2	213	37.2	1.93	460.2	19.5	1.3
		2		3.3	135	32.2	1.67	548.9	15.6	1.1
		0.5	200	3.8	312	24.8	2.57	547.6	9.7	1.7
100	2	15	30	4	138 ± 3	82.6 ± 2.1	1.28 ± 0.03	259.6 ± 6.5	34.3 ± 0.9	0.9 ± 0.1
		15	100	5.6	412 ± 14	74.6 ± 2.4	3.86 ± 0.12	402.6 ± 13.4	22.1 ± 0.7	2.6 ± 0.1
		15	200	6.4	756 ± 21	67.6 ± 2.8	7.00 ± 0.28	507.7 ± 12.1	17.5 ± 0.7	4.7 ± 0.2
		15	300	7	353 ± 85	21.0 ± 5.8	3.27 ± 0.57	1784.7 ± 151	5.0 ± 0.8	2.2 ± 0.2
		7	200	6.7	680 ± 45	28.3 ± 1.9	2.94 ± 0.2	1266.8 ± 84	7.0 ± 0.5	2.0 ± 0.1
		20	200	5.6	350 ± 20	41.7 ± 2.4	4.32 ± 0.3	720.0 ± 41	12.4 ± 0.7	2.9 ± 0.2

due to enhancements in CO₂ distribution and reaction design. This confirms the technical feasibility of scaling up the gas-phase CO₂ electroreduction to formate using a zero-gap GDE configuration. However, additional challenges must be addressed for industrial deployment. Notably, formate output must meet industrial-scale requirements, necessitating either an increase in single-cell area or the development of multicell stack systems. Furthermore, reducing EC and minimizing internal cell resistances are critical to improving process viability. Therefore, this work represents an important step toward the industrial demonstration of CO₂-to-formate electroreduction. The performance gains achieved at the prototype scale underline the potential of this technology to become a competitive solution for CO₂ valorization.

4. Conclusion

The scaling-up of CO₂ electroreduction to formate is still in the early stages of research, as only a limited number of studies have addressed this aspect. More specifically, the scale-up of gas-phase CO₂ electroreduction to formate using a zero-gap configuration at the cathode GDE has barely been explored. For this reason, this work focuses on the design of a novel 100 cm² electrolyzer prototype, to demonstrate the scalability of the process prior its industrial validation.

Scaling up a CO₂ electrolyzer to a 100 cm² prototype introduces several challenges, particularly the electrode fabrication and integration in the electrolyzer. In this case, the automated fabrication by spray pyrolysis enables the reproducible scaling up of the GDE preparation. Besides, the GDE is integrated into the electrolyzer employing a serpentine flow field to provide the CO₂ to its surface, ensuring good CO₂ concentration management within the reaction area, which results in favoring the CO₂ mass transfer.

The prototype is tested under varying operational parameters, including current density and water content in the CO₂ feed stream, to assess its performance. The experimental evaluation of the 100 cm² electrolyzer prototype for the gas-phase conversion of CO₂ to formate enabled the identification of optimal operating conditions and the main limiting factors of the process at prototype scale.

First, increasing the current density improves the reaction kinetics, with a maximum performance observed at 200 mA cm⁻². Under these conditions, a formate concentration of 760 g L⁻¹, FE of 67%, and a formate rate of 7 mmol m⁻² s⁻¹ are achieved. However, operating at higher current densities, such as 300 mA cm⁻², led to reduced performance due to phenomena associated with GDE electrowetting, which limits CO₂ access to the catalyst. As a result, the formate production rate drops to 3.27 mmol m⁻² s⁻¹, with FE and concentration decreasing to 21% and 353 g L⁻¹, respectively, well below the values obtained at 200 mA cm⁻².

Regarding the water content in the CO₂ feed, an optimal condition is also identified that balances the need to provide sufficient humidity for the reaction and prevent membrane drying, while avoiding excess water that could flood the GDE and hinder

CO₂ mass transfer. This balance is achieved with a water flow rate of 15 g h⁻¹ in the CO₂ stream, which maximized both the formate concentration (760 g L⁻¹) and the system's EE (17.5%). Lower water input (7 g h⁻¹) resulted in membrane dehydration (680 g L⁻¹ and 7% EE), while higher input (20 g h⁻¹) led to electrode flooding (350 g L⁻¹ and 12.4% EE).

After analyzing the prototype's behavior under different reaction conditions, its performance is compared with that of a laboratory-scale system. Overall, an improvement in key performance metrics is observed, partly due to enhanced CO₂ distribution via the serpentine flow field. More than double the formate concentration is achieved under comparable conditions (from 312 to 756 g L⁻¹), along with a 40% increase in FE and a rise in SPCE from 1.7% to 4.7%. However, scaling up resulted in increased Ohmic resistance, attributable to both the larger electrode size^[48] and to the challenges in managing bubble formation at the anode, reflected in a higher cell voltage, from 3.8 to 6.4 V, which limited improvements in EC compared to other metrics. This increased resistance remains one of the key challenges that must be addressed to enable a more competitive scale-up of the technology.

Furthermore, increasing the reaction area is necessary to enhance the system's daily formate production capacity, either by developing larger electrolyzers or by implementing cell stacks. Therefore, considering this as a first step toward the industrial demonstration of gas-phase CO₂ electroreduction to formate, further research is essential to improve the technology. The results presented here demonstrate its competitiveness as a viable alternative for utilizing captured CO₂ to produce value-added products, specifically, formate, and bring the technology closer to industrial implementation. However, future research efforts should focus on studying the behavior of the system under longer operating times, as deactivation phenomena, such as those mentioned above, including salt precipitation within the GDE or electrode flooding, can negatively impact the scalability of the technology. Therefore, their effects and potential mitigation strategies should be thoroughly investigated.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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